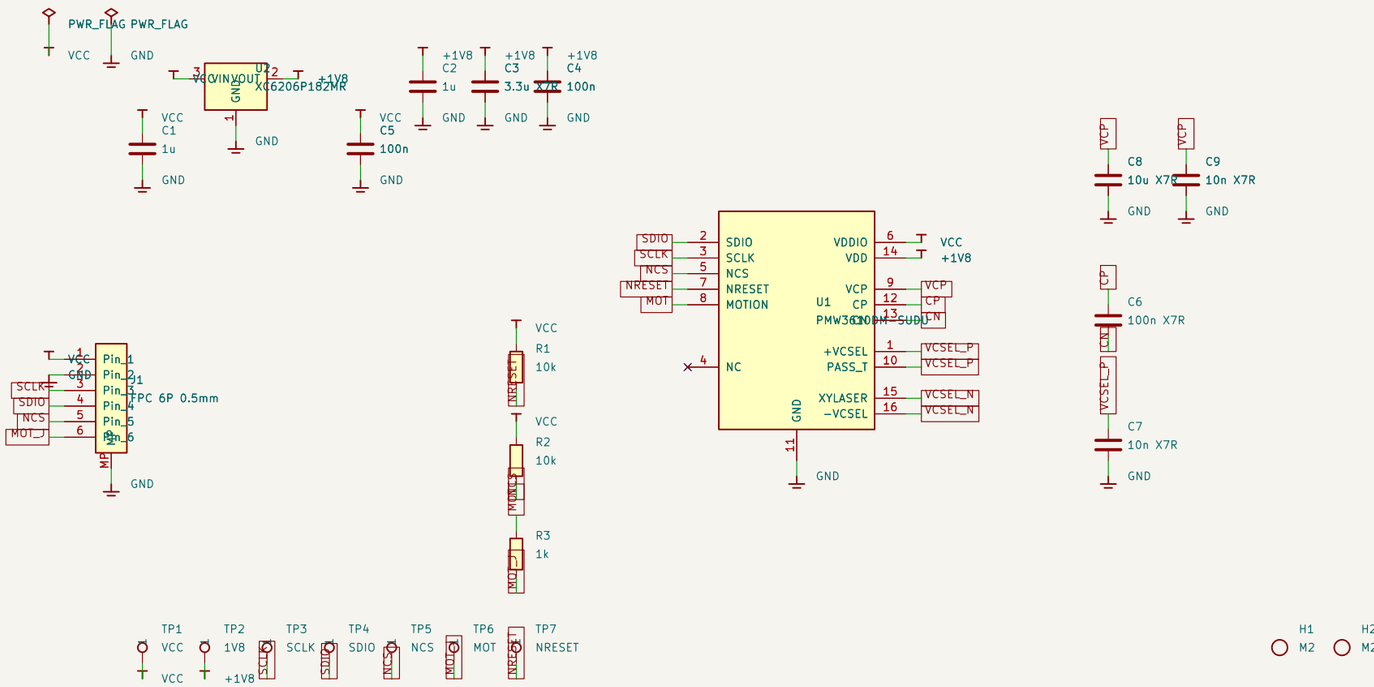


k0d3ball35 trackball sensor board, PMW3610DM−SUDU + LM18−LSI lens.
VDDIO and all logic run at 3.3 V straight from the main PCB (no level shifters).
VDD 1.8 V is generated on board by U2. MOTION is push−pull; R3 limits current if the
FFC is inserted the wrong way round. R2 keeps the sensor deselected while the host boots.

FPC pin order on THIS board: 1 VCC, 2 GND, 3 SCLK, 4 SDIO, 5 NCS, 6 MOT.
Same order as the main PCB: J1 sits on the KEY side there, so its pads are not mirrored.
Both connectors are bottom−contact parts; the cable is a type B FFC (contacts on
OPPOSITE faces at its two ends). A type A cable will not work.

FPC pin order is mirrored relative to the main PCB; see docs/08−trackball−module.md

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k0d3ball35 project

Sheet: /
File: k0d3ball35−sensor.kicad_sch

Title: k0d3ball35 trackball sensor board (PMW3610)

Size: A3	Date: 2026−11−05	**Rev: r1**
KiCad E.D.A. 10.0.7		Id: 1/1